# Robust Aerial Person Detection With Lightweight Distillation Network for Edge Deployment

Xiangqing Zhang, *Graduate Student Member, IEEE*, Yan Feng, *Member, IEEE*, Shun Zhang, *Member, IEEE*, Nan Wang, *Graduate Student Member, IEEE*, Guohua Lu, and Shaohui Mei, *Senior Member, IEEE*

***Abstract*— Aerial person detection (APD) is vital for enhancing search and rescue (SaR) operations, particularly when locating victims in remote, poorly-lit areas. Despite advancements in detection technologies, achieving a balance between detection speed and accuracy on mobile devices in "edge AI" continues to pose challenges. In this article, a lightweight distillation network (APDNet) is proposed for edge deployment of APD, which enables real-time inference as well as minimizes accuracy loss during model transfer. The proposed APDNet employs a distillation network between varying-depth backbones and integrates an 8-bit quantized optimizer to reduce the floating-point operations of network parameters. Specifically, in the teach-assistant distillation (TAD) stage, small student models using random weight initialization are trained with pseudo-labels generated by deeper teacher models, facilitating consistent learning for a more accurate, lighter model. Moreover, a low-precision quantization (LPQ) stage incorporates an offline, quantization-aware training strategy that dynamically adjusts the ranges of weight and activation function float-point values, reducing computational complexity. In order to compensate for the potential accuracy decline, a pluggable tracker updates the position and feature information of persons frame-by-frame, with tracking results integrated with detection outputs to enhance accuracy. Extensive experiments on the Heridal, Manipal-UAV, and VTSaR datasets confirm the effectiveness of APDNet, demonstrating its superior performance in edge-based APD.**

***Index Terms*— Aerial person detection (APD), distillation network, pluggable tracker, quantization awareness.**

## I. Introduction

AERIAL person detection (APD) is crucial for intelligence surveillance and search and rescue (SaR) operations, fueling extensive research in numerous countries [1], [2], [3], [4], [5], [6], [7], [8]. This task aims to detect persons' positions in images captured from unmanned aerial vehicles (UAVs). Researchers have developed various detection pipelines inspired by Glenn Jocher's groundbreaking work [9] to detect efficiently. These pipelines consist of object detection in natural images [10], [11], person detection in aerial images [12], and model acceleration with lightweight detectors [13], [14], which strive to enhance both the accuracy and computational efficiency of the detection system. However, challenges arise due to the distinctive characteristics of high-resolution aerial images, such as a wide field of view, significant angle variations, and small object sizes. These challenges result in problems like imprecise aerial-based person detectors (APDs) and obstacles in implementing real-time, low-power detection algorithms. Within this framework, as shown in Fig. 1, the APD task can be subdivided into two subtasks: lightweight deployment models and robust detectors. The former aims to fulfill real-world application requirements on edge computing. At the same time, the latter focuses primarily on extracting accurate personal motion information from sequences of images given the context of the lightweight detection model.

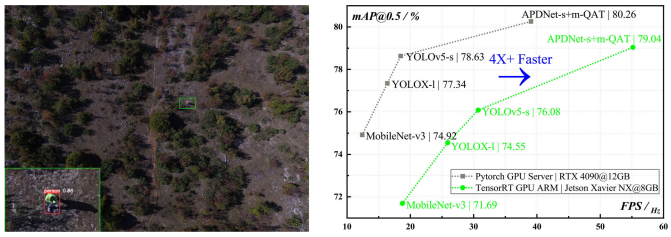



Fig. 1. (Left) Exemplar of person detection results for APDNet (Ours). (Right) Comparing performance curves between SOTA methods and APDNet (Ours) under GPU server/ARM devices.

Recently, numerous deep learning-based AI methods have emerged. For instance, Hong et al. [15] proposed a highly efficient base model, "SpectralGPT," demonstrating excellent accuracy on various datasets, introducing new approaches and thoughts in generative AI for remote sensing. LRR-Net [16] introduced a sparse neural network for hyperspectral anomaly detection, showcasing the scalability of the LRR-Net framework. Li et al. [17] put forward a method that merges CASSI and supplementary measurements, successfully applied to the blind reconstruction of complex images. Similarly, substantial progress has been made in the field of visual object detection [18], [19], [20], [21]. However, the majority

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Xiangqing Zhang is with the School of Electronics and Information, Northwestern Polytechnical University, Xi'an 710129, China, and also with the College of Mathematics and Computer Science, Yan'an University, Yan'an 716000, China (e-mail: zxq1991@mail.nwpu.edu.cn).

Yan Feng, Shun Zhang, Nan Wang, and Shaohui Mei are with the School of Electronics and Information, Northwestern Polytechnical University, Xi'an 710129, China (e-mail: sycfy@nwpu.edu.cn; szhang@nwpu.edu.cn; wangnan418@mail.nwpu.edu.cn; meish@nwpu.edu.cn).

Guohua Lu is with the Department of Military Biomedical Engineering, Air Force Medical University, Xi'an, Shaanxi 710032, China (e-mail: lugh1976@fmmu.edu.cn).

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of deep learning methods for object detection have been developed for medium to large objects, and their performances on small-object detection (for example, tiny persons less than 20 pixels) in large-scale images remain not well investigated (APD constitutes a distinctive scenario within the realm of small-object detection) [22]. As shown in Fig. 2, there are two challenges from aerial person detectors: the limited inference speed for edge computing indicates that a person is detected with constrained edge hardware, and the difficulty locating looming persons means that a background region is mistakenly detected as a person. These two challenges are elaborated as follows.

The limited inference speed for edge computing is triggered by high-precision convolutional neural networks (CNNs) models, which typically possess deeper network structures combined with high-resolution input images. As shown in Fig. 2(a), a positive correlation exists between the size of a model's parameters and the inference latency. Recently, the runtime inference in the person detection task has attracted much more attention from researchers. Xu et al. [23] proposed an extremely lightweight DNN with 12 convolutional (Conv) layers and only 1.82 MB of parameters following a bottom-up DNN design approach. Shamsolmoali et al. [24] proposed the rotation equivariant feature image pyramid network (REFIPN) based on rotation equivariance convolution. Nasser et al. [25] envision a privacy-preserving pandemic response network using a proof-of-concept, aerial-terrestrial network system serving mobile user entities/equipment (UEs). Yang et al. [26] proposed an adaptive reinforcement supervision distillation (ARSD) framework to promote the detection capability of the lightweight model. Aiming at the difficulties in RSIs, Zhou et al. [27] proposed an object detection method based on the dense feature fusion path aggregation network (DFF-PANet). Li et al. [28] proposed an SAR ship detection method with ultralightweight and high detection accuracy based on YOLOX. However, incorporating lightweight methods into detection tasks without assessing the potential accuracy loss can often lead to adverse effects, underscoring the necessity for further investigation.

The nonrobust classification ability leads to difficulty in detecting approaching persons, resulting in numerous false alarms and missed true positives. As shown in Fig. 2(b), person detection can be readily confounded by background features such as trees, mountains, or seawater. Of late, the precision associated with person detection has garnered substantial scholarly interest. Jing et al. [29] contributed to research in low-resolution small-object detection by evaluating the performances of leading deep learning methods for object detection using a standard dataset created from real-life aerial imagery data. Yu et al. [30] introduced a new benchmark called TinyPerson, which opens up a promising direction for detecting tiny objects at long distances and within extensive backgrounds. Kundid Vasić and Papić [31] proposed a novel method for person detection in aerial images of nonurban terrain gathered by a UAV, which plays a vital role in SaR missions. To advance tiny object detection research in aerial images, Wang et al. [32] presented a new dataset for tiny object detection in aerial images (AI-TOD). Zhang et al. [33] introduced the image cropping strategy and attention mechanism based on YOLOv5 to address small person detection in the optimized VisDrone2019 dataset. Wang et al. [34] studied person detection and identification surrounding gas facilities based on improved-yolov4-tiny. However, these above methods concentrate solely on mitigating the issue of feature information loss associated with small target detection, overlooking the influences of background interference and contextual information. This oversight has resulted in a significant deterioration in detection accuracy.

To address the noted challenges, we propose a lightweight method called APDNet, featuring three core components: teach-assistant distillation (TAD), low-precision quantization (LPQ), and a robust detection module (RDM). These are designed to enhance the performance of aerial person detectors. In the TAD component, small student models employ random weight initialization and use response-based pseudo-labels from deep teacher models for retraining within a consistency loss. The LPQ component utilizes an offline, quantization-aware training strategy that adaptively maps floating-point values of weights and activation functions to optimized intervals, significantly reducing computational complexity. In addition, the RDM component incorporates a plug-in tracking module that continuously updates the positional and feature information of objects across frames. The integration of these tracking results with detector outputs enhances object detection accuracy, establishing APDNet as a new state-of-the-art method in APD. Comprehensive experiments on the "Heridal [35]," "Manipal-UAV Person detection (MUPD) dataset [36]," and "VTSaR" datasets served to validate the efficacy of our APDNet methodology, demonstrating cutting-edge performance in APD on edge platforms. The main contributions of this article can be summarized below.

1) Proposing an APDNet framework to conduct a distillation model between backbones of diverse depths, integrating an 8-bit quantized optimizer to diminish the floating-point computations of network parameters. Furthermore, the framework incorporates a tracker-assisted detection feature to counteract any potential decrease in accuracy associated with the model's lightweight design, which effectively enhances the accuracy of APD while facilitating model deployment on edge devices.
2) Introducing a TAD module allows student models to acquire and transfer logical reasoning capacities similar to those exhibited by teacher models with response-based distillation and consistency loss, which effectively diminishes the model's computational complexity.
3) Designing an LPQ module that maps data onto a dynamically constructed tree, selectively quantizing nonuniformly distributed parameters and isolating outliers, typically represented by maximum and minimum values, minimizes information loss during activation quantization and effectively accelerates inference speed.
4) Constructing an RDM module incorporates plug-and-play trackers that provide feedback to detectors. This setup helps mitigate accuracy declines due to person

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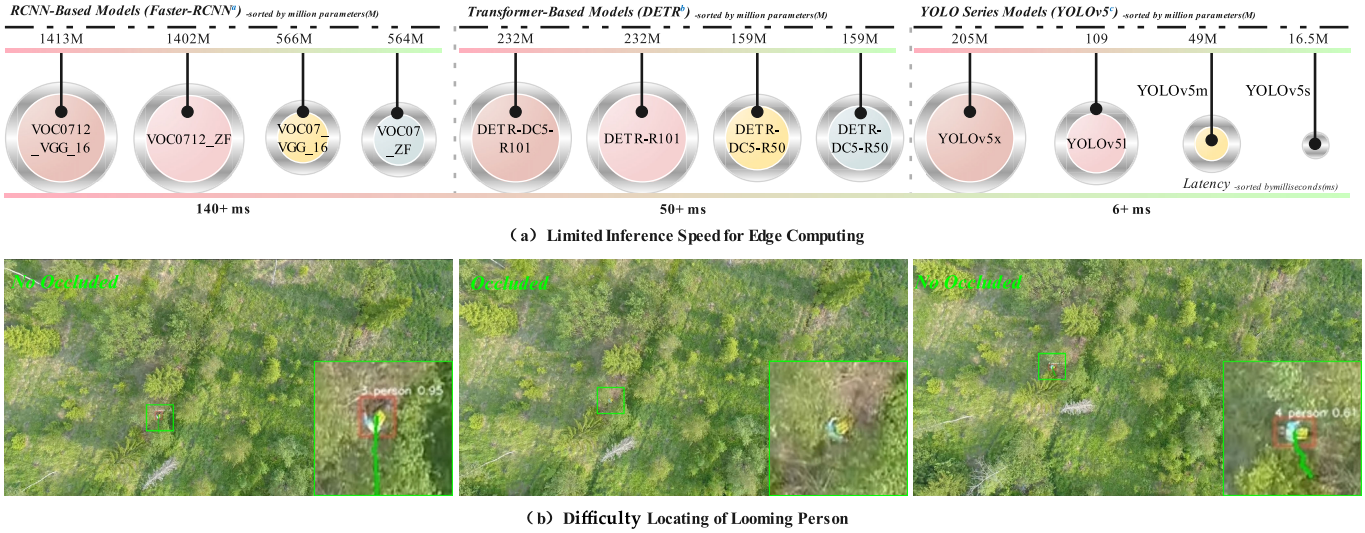



Fig. 2. Challenges exist in the deployment of detection models. (a) High-precision CNN models typically require deeper network layer connections and entail substantial storage or power consumption. (b) Aerial-vision-based objects of persons are susceptible to background occlusion and interference.

occlusion in SaR scenarios and compensates for the reduced accuracy of lightweight detectors.

The remainder of this article is structured as follows. Section II discusses the challenges drawn from related works. Section III elaborates on the detailed design of the proposed method. The experimental validation of our approach is presented in Section IV. Finally, Section V summarizes the article and highlights potential future directions.

## II. Related Works

This section provides an overview and analysis of previous research related to the topic. Subsequently, the challenges associated with existing methods will be outlined, and the proposed ideas to address these issues will be presented.

### A. Lightweight Object Detection

Lightweight object detection methods aim to reduce model complexity and improve runtime efficiency in resource-constrained environments. Several common lightweight object detection methods mainly include model compression [37], network pruning [38], LPQ [39], lightweight network [40], and model distillation [41]. These methods can be applied individually or in combination to achieve efficient object detection in resource-constrained environments.

Past studies have predominantly focused on enhancing the performance of an ideal detector in a singular aspect, disregarding the need to strike a balance between accuracy and speed. It is crucial to find an appropriate tradeoff to ensure detection accuracy aligns with processing speed. Therefore, we propose an integrated approach that employs existing single-stage detectors. Leveraging feature mimicry across models of varying depths, it incorporates quantization techniques, including TAD, LPQ, and RDM modules, to effectively augment detection performance.

### B. Knowledge Distillation

Knowledge distillation is a technique that enhances performance by transferring knowledge from a complex model (teacher) to a simplified (student) model [42]. Standard knowledge distillation methods include response-based [43], [44], which utilizes the teacher model's classification predictions as the "target knowledge" for the student model; feature-based [45] refers to the student model learning from the features extracted by the teacher model; relation-based [46] focuses on the relationships between samples or feature layers. These methods offer distinct approaches to knowledge transfer, enabling the student model to benefit from the expertise of the teacher model.

In comparison, response-based knowledge distillation avoids the need for modifying complex network structures and demonstrates extensive applicability. Nonetheless, it faces challenges related to the incomplete transmission of knowledge from the teacher model to the student model, which can result in potential knowledge loss or degradation. Furthermore, soft labels may introduce additional noise and errors, impeding the training of the student model. To address these issues, this study adopts soft labels in the form of probability distributions from the teacher model, departing from the conventional use of hard labels. In addition, we leverage pseudo-labels generated by the teacher model as auxiliary information to guide the training of the student model.

### C. Quantization Strategy

Model quantization decreases the precision of parameters and activations in large models, converting them from high-precision to low-precision representation [39]. Standard methods of model quantization include: weight quantization [47] compresses the weight parameters of the model by converting them from floating-point representation to lower-bit-width integers, such as binary quantization, ternary quantization, and multibit quantization. Activation

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quantization [47] is a technique that converts the activation values in a neural network into lower-precision representations, such as symmetric and asymmetric quantization. Network pruning and quantization [48] involve removing redundant connections and neurons, compressing weight parameters, and quantizing the floating-point weights into lower-bit-width integers.

While quantization methods have advanced considerably, they still face challenges, including introducing quantization errors during low-precision representation, the inability to fully recover the source model information after quantization, and the potential increase in training complexity when integrating quantization. Accordingly, we utilize classical single-stage detector models as exemplars for quantization by carefully selecting quantization strategies, refining hyperparameters, and optimizing training procedures to achieve an 8-bit quantization with reduced precision.

## III. Methodology

This section comprehensively describes the APDNet and three modules for lightweight APD.

### A. Overview of APDNet

Deployment of deep learning-based detection models encounters critical challenges related to high-precision CNN models that often require deeper network layer connections and numerous storage or power consumption by wide-bit floating-point parameters. To address these challenges, this study introduces a method that transfers knowledge from a complex teacher model to a smaller, simplified student model, thereby reducing the number of parameters. In addition, using a lower bit representation for floating-point numbers achieves the same value range, reducing storage costs. Fig. 3 illustrates the three main modules of our approach, namely, APDNet: TAD, LPQ, and RDM. The TAD module computes the intersection over union (IoU) between the predicted bounding boxes of the teacher and student models, adopting a consistent IoU as the loss function for the detector. In the LPQ module, a variable structure tree model is introduced to encode the weights and biases of dynamically connected nodes, enabling adjustments in branching mechanisms and the addition/removal of nodes. The RDM module exploits object motion information in video sequences to estimate potential object locations, effectively addressing occlusion and background changes. Detailed descriptions of each module will be provided in Sections III-B–III-D.

### B. Slighter Model With Logit Mimicking

*1) Teach-Assistant for Response-Based Distillation:* Response-based knowledge distillation is a technique used to transfer knowledge from a large pretrained model to a smaller model while ensuring the smaller model maintains high-performance efficiency. This approach involves both the teacher model and the student model receiving the same input data and generating corresponding prediction responses. Fig. 4 illustrates the utilization of YOLOv5, a deeper pretrained model, to accurately represent the features of the sample data and treat it as knowledge from the original model. Subsequently, the final detection model is trained by minimizing the disparity between the outputs of the original model and the smaller model. As shown in Fig. 4, the top network represents the teacher network and the bottom network represents the student network. In order to transfer the knowledge from the teacher network to the student network, where P3–P7 achieves rich feature information of objects from the backbone and neck, an improved KD divergence loss is constructed in this article.

*2) Consistency Loss:* Consistency loss evaluation is achieved by aligning the outputs of all models at the final activation layer. By employing this method, the smaller model is capable of acquiring the extensive knowledge present in the larger model. During the process of response-based knowledge distillation, the normalized probabilities, known as soft labels, are defined as $q_i$ and obtained through the softmax function

$$q_i = \frac{\exp(z_i/T)}{\sum_j \exp(z_j/T)} \tag{1}$$

where $z_i$ represents the unnormalized probabilities known as logits, and $T$ denotes the distillation temperature. As the temperature $T$ increases, the probability distribution of each class becomes smoother. The response-based distillation loss ($L_{\mathrm{KD}}$) is defined as the weighted sum of two components, as shown below

$$L_{\mathrm{KD}} = \alpha L_{\mathrm{soft}} + \beta L_{\mathrm{hard}} \tag{2}$$

where $\alpha$ and $\beta$ are soft values according to the training stages of teacher and student models. The first component $L_{\mathrm{soft}}$ measures the consistency between the soft labels generated by the teacher model and the student model, which can be expressed as

$$L_{\mathrm{soft}} = -\sum_i^N \frac{\exp(v_i/T)}{\sum_k^N \exp(v_k/T)} \log\left(\frac{\exp(z_i/T)}{\sum_k^N \exp(z_k/T)}\right) \tag{3}$$

where $v_i$ is the logits of the teacher; $z_i$ is the logits of the student; and $N$ is the number of input instances. The second component, $L_{\mathrm{hard}}$, measures the consistency between the predictions of the student model and the true labels. This can be expressed as follows:

$$L_{\mathrm{hard}} = -\sum_j^N c_j \log\left(\frac{\exp(z_j)}{\sum_k^N \exp(z_k)}\right). \tag{4}$$

In this case, $c_j$ represents the actual label of instance $j$, where $c_j \in 0, 1$ (with a value of 1 assigned to positive labels and a value of 0 assigned to negative labels).

*3) Loss Function:* The loss function of our proposed method consists of three components: the classification loss ($L_{\mathrm{cls}}$), confidence loss ($L_{\mathrm{obj}}$), and bounding box regression loss for object detection ($L_{\mathrm{bbox}}$), which are similar to those used in YOLOv5. Finally, incorporate the distillation loss ($L_{\mathrm{KD}}$) to assess the distillation divergence between teacher and student models.

The classification loss ($L_{\mathrm{cLs}}$) is calculated using the binary cross-entropy function, which is defined as

$$L_{\mathrm{cLs}} = L_{\mathrm{BCE}}$$

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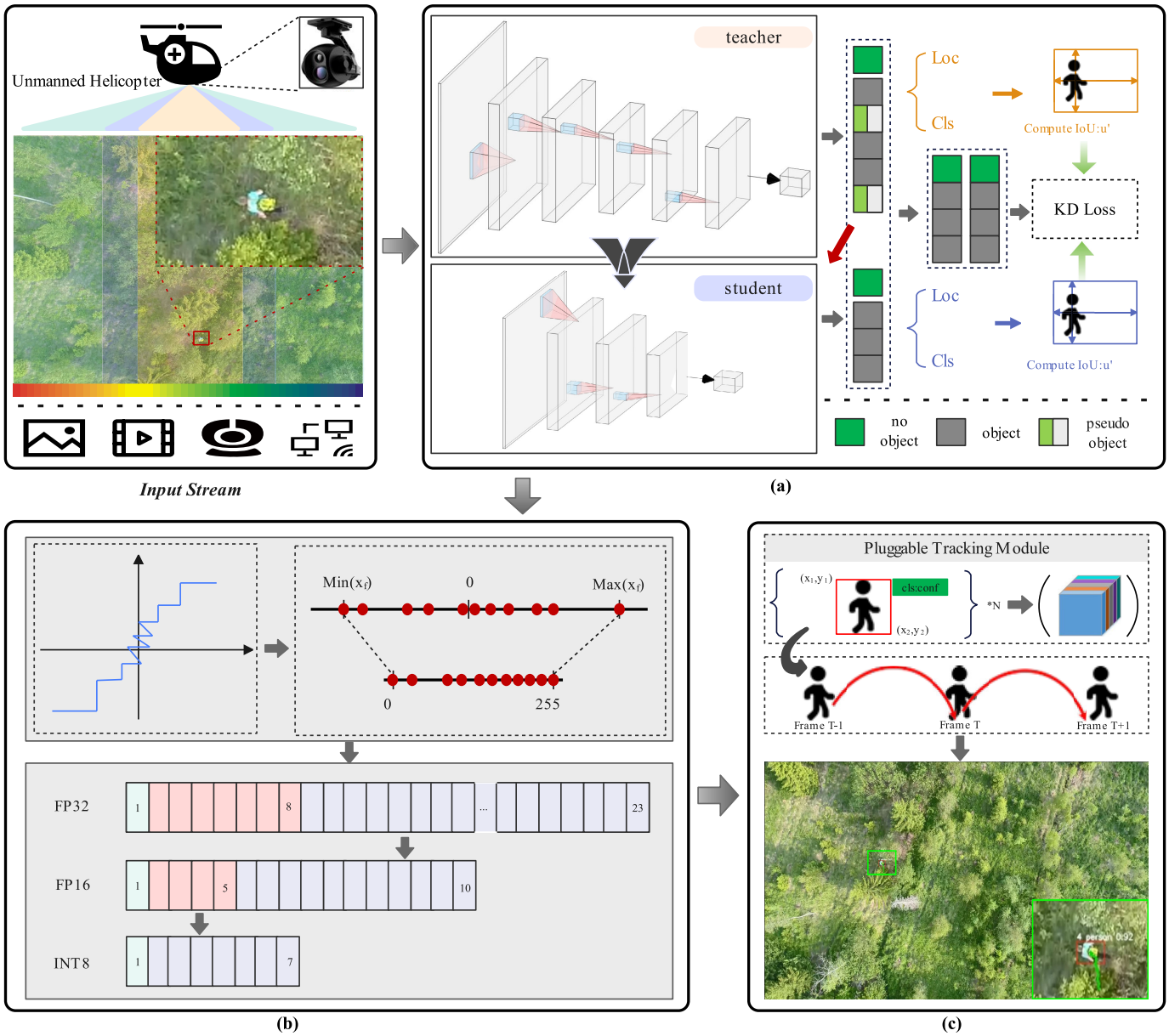



Fig. 3. Framework of APDNet, which consists of three modules: (a) TAD, which primarily transfers the well-learned training parameters from the teacher to the lighter student model, and obtains the location distillation and classification probability based on the loss of predicted labels and the GTs. (b) LPQ employs a dynamic tree structure-based quantization strategy to reduce 32-bit-width floating-point representations to 8-bit integers of weights and biases, thereby minimizing computational resources overhead. (c) RDM integrates object detection and tracking methods. By feeding back the tracking prediction results to the detection inference, detection accuracy is further improved.

$$= -y \log p - (1-y)\log(1-p)$$
$$= \begin{cases} -\log p, & y = 1 \\ -\log(1-p), & y = 0. \end{cases} \tag{5}$$

In this context, the label for the input sample is denoted by $y$, where a value of 1 indicates a positive sample and 0 represents a negative sample. In addition, the predicted probability assigned by the model for classifying the input sample as positive is represented by $p$.

The confidence loss ($L_{\text{obj}}$) is calculated using the binary cross-entropy function and incorporates weight coefficients that vary based on the detection layers' scales. These coefficients are assigned in descending order of the detection layer's scale, with default values of 4.0, 1.0, and 0.4. Consequently, small objects receive higher weight coefficients for the loss on multiscale feature maps, thus emphasizing the training focus on small objects

$$L_{\text{obj}} = L_{\text{cls}}. \tag{6}$$

The bounding box regression loss ($L_{\text{bbox}}$) addresses the dissimilarities between predicted bounding boxes and ground truth (GT), CloU takes into account additional factors such as aspect ratio. To mitigate this issue, CloU introduces two new parameters

$$L_{\text{bbox}} = L_{\text{CloU}(B,B_{\text{gt}})} = \text{IoU}(B, B_{\text{gt}}) - \frac{\rho^2(B, B_{\text{gt}})}{c^2} - \alpha v$$

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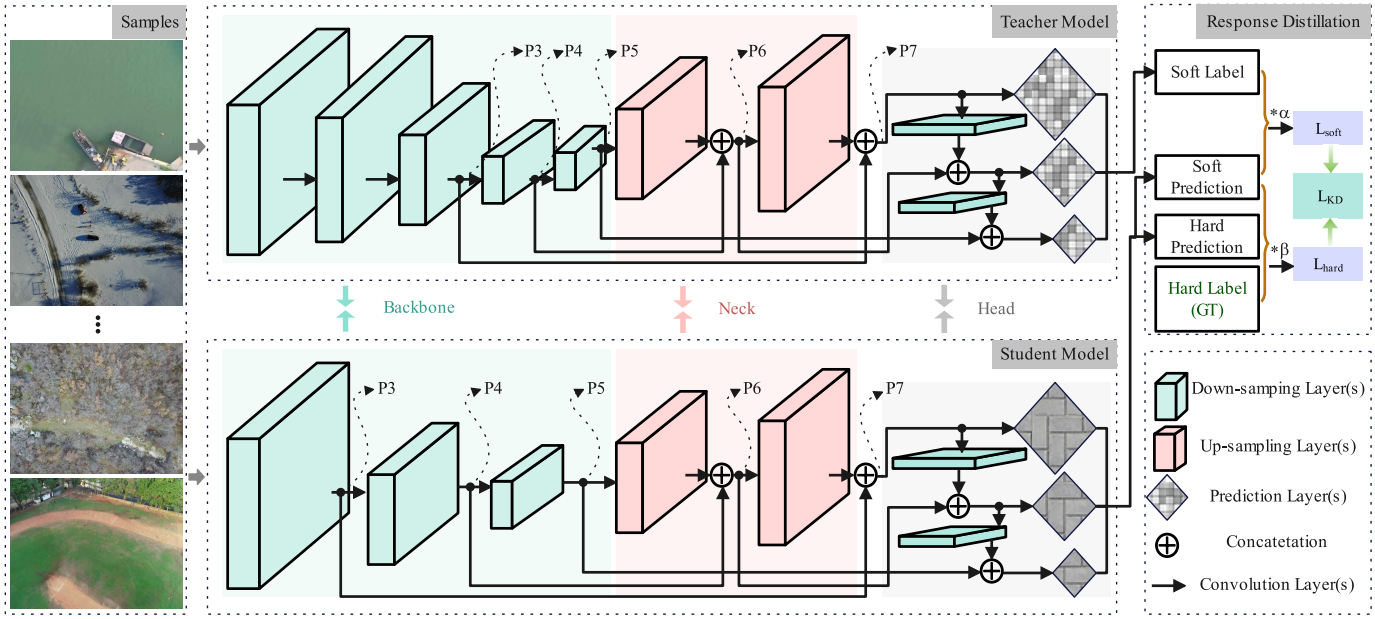



Fig. 4. Architecture of TAD module. The symbols P3–P7 represent the levels of the feature pyramid in YOLOv5. P3 is a high-resolution feature map outputted from the backbone network. P4–P7 represent feature maps with progressively reduced resolution, where downsampling operations are employed. $\alpha$ and $\beta$ denote the weight factors used to compute the distillation loss for the teacher and student, respectively.

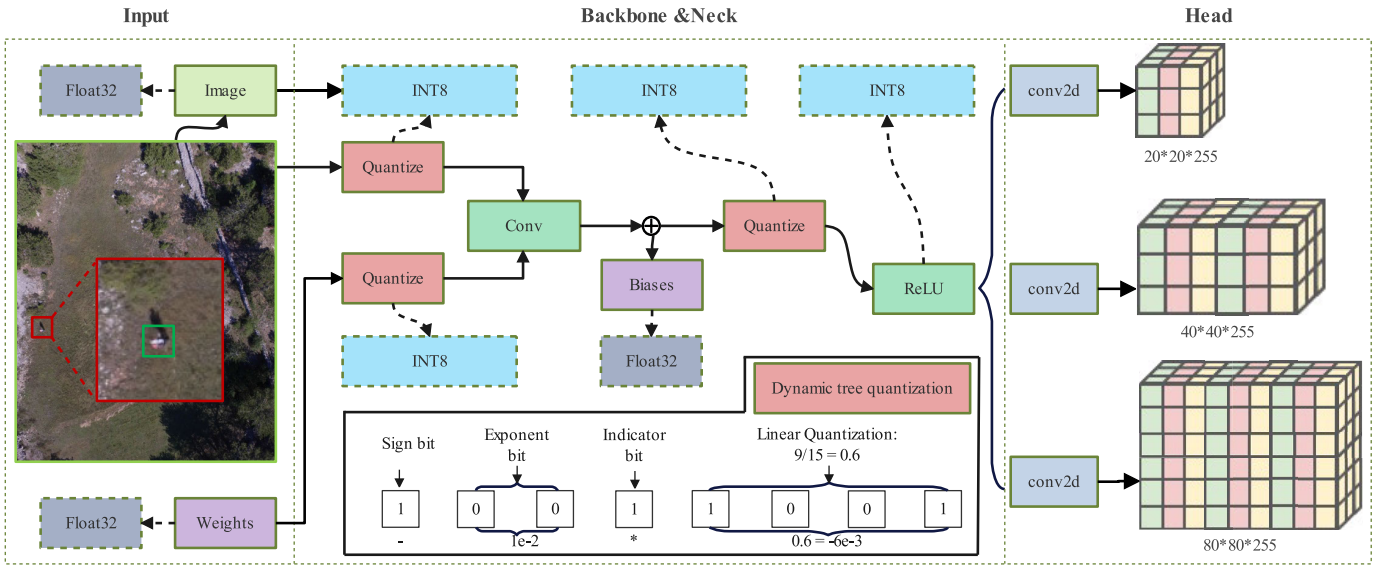



Fig. 5. Architecture of LPQ module.

$$v = \frac{4}{\pi}\left(\arctan\frac{w^{\mathrm{gt}}}{h^{\mathrm{gt}}} - \arctan\frac{w}{h}\right)^2$$
$$\alpha = \frac{v}{1 - \mathrm{IoU}(B, B_{gt}) + v}. \tag{7}$$

The parameter $v$ denotes the normalized difference in aspect ratio between the predicted and GT. The parameter $a$ ranges from 0 to $\pi/4$ and is subsequently multiplied by $\pi/4$ to ensure it falls within the range of 0–1. In addition, the parameter $\alpha$ is a balancing factor for the tradeoff between the loss resulting from the aspect ratio difference and the loss resulting from IoU.

The overall loss ($L_{\mathrm{Total}}$) should be expressed as follows:

$$L_{\mathrm{Total}} = L_{\mathrm{cls}} + L_{\mathrm{obj}} + L_{\mathrm{bbox}} + L_{\mathrm{KD}}. \tag{8}$$

### C. Block-Wise Quantization at 8-bit Optimizers

To mitigate the substantial memory overhead caused by high-precision floating-point parameters, this section presents a dynamic tree quantization method [47] as illustrated in Fig. 5. We employ block-wise quantization to the weights and biases of the backbone, neck, and head modules in the YOLOv5 framework. By selectively quantizing nonuniformly distributed parameters and isolating outliers, typically represented by the maximum and minimum values, the information loss during the activation quantization process is minimized, thus ensuring precise object detection results. Finally, the fully quantized network is deployed on GPU-equipped edge devices, striking a balance between maintaining a reasonable level of accuracy and enabling efficient inference.

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In LPQ, defining an appropriate quantization range is essential to ensure accurate results. This is accomplished by setting upper ($u$) and lower ($l$) threshold values for the quantization grid to strike a balance between truncation and rounding errors (as represented in (9) and (10), respectively). Truncation error is commonly observed when data is truncated to fit predefined grid boundaries, leading to information loss and a reduction in the accuracy of the resulting quantized representation. On the contrary, rounding error originates from the inherent imprecision introduced during the rounding process

$$s_{\mathbf{x}} = \frac{\mathrm{u}-1}{2^b-1}, \mathrm{zp}_{\mathbf{x}} = \mathrm{clip}\left(\lfloor -\frac{l}{s} \rceil, 0, 2^b-1\right) \tag{9}$$

$$\mathrm{Q}(\mathbf{x}|b) = \mathrm{clip}\left(\lfloor \frac{\mathbf{x}}{s_{\mathbf{x}}} \rceil + zp_{\mathbf{x}}, 0, 2^b-1\right) \tag{10}$$

where the quantization bit width is denoted as $b$. The quantizer $Q(\mathbf{x}|b)$ is defined as a function that maps a floating-point number, $x \in R$, to the closest quantization bin. The quantization parameters, $s_{\mathbf{x}}$ (scale) and $zp_{\mathbf{x}}$ (zero point), are introduced. The symbols $u$ (upper) and $l$ (lower) are utilized to specify the limits of the quantization grid.

Adopting the block-wise quantization method with dynamic trees improves absolute and relative quantization errors. Specifically, the floating-point data is partitioned into four components: dynamic exponent and fraction, as demonstrated in Fig. 5 with the dynamic tree quantization module. The sign is assigned to the first bit, while the subsequent bits, varying in number, indicate the exponent's magnitude. The fourth bit is set as one by default, and the remaining bits undergo linear quantization. Through sequential shifting of the block-wise indicator bit, high-precision floating-point values are successfully quantized into 8-bit integers. It is worth noting that all tensors are normalized to the range of $[-1, 1]$. The quantizers are applied to each weight front end of the detector convolutional layer separately to ensure a reasonable truncation of the floating-point bits, and the quantizers are also applied to the activation layer to make the output weights/biases tails as low as possible.

### D. Robust Detection With Pluggable Tracker Modules

In the image-based object detection paradigm, persons frequently experience occlusion or interference caused by significant or moderate-sized background objects such as forests, mountains, or bodies of water. Consequently, detection results become inconsistent across different video frames from aerial-vision-based devices. To handle this issue, this section employs ensemble learning to combine the motion information from a tracker with the spatial information from a detector. The main objective is to mitigate the reduced accuracy of the detector caused by false positives or misclassifications in the current frame. As is shown in Fig. 3(c), the detector quickly identifies candidate objects in various temporal frames and forward the detected candidate boxes to the tracker. The tracker then assesses the IoU overlap between consecutive frames to determine whether to predict the object's position in the current frame. Using a voting mechanism, if the IoU calculation between the detector's output and the GT exceeds that between the tracker's predicted value and the GT, the detector's output is selected as the algorithm's final result. Conversely, if the detector fails to identify the target in the current frame, the tracker's predicted value serves as the algorithm's ultimate output. This approach culminates in an improved inference of the object's position and orientation, thus enhancing the overall accuracy of the detection process.

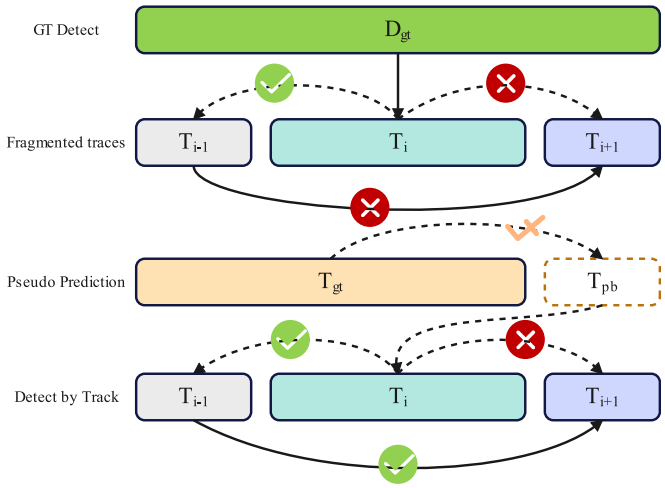



Fig. 6. Explanation of tracking-assisted detection.

Tracking-assisted detection is carried out through a two-stage execution process. Initially, an effective object-tracking algorithm was utilized to track the objects within the video sequence and to gather their positional and motion information. Subsequently, these tracking results were integrated with the object detector, as illustrated in Fig. 6, $D_{\mathrm{gt}}$ is the detector's GT. When no object is detected in the current frame, denoted as $T_i$, the tracking algorithm predicts the target's position in the subsequent frame, $T_{i+1}$. The predicted bounding box ($T_{pb}$) was then compared to the bounding box of the previously identified target in frame $T_{i-1}$, using the IoU calculation. If the overlap fell within a predetermined threshold, the target's bounding box at frame $T_{i+1}$ was shifted to align with the current frame $T_i$, thereby facilitating the detection of the object and enhancing the detector's accuracy.

## IV. Experiments

This section discusses the experimental setup, including data collection, preprocessing, and the chosen evaluation metrics. The experiments are designed to validate the proposed method's effectiveness and performance and compare it with existing approaches.

A comprehensive comparison of various metrics commonly employed in the training stage, including precision, recall, and mean average precision (mAP). Each metric is examined in terms of its strengths, limitations, and suitability for different detection scenarios. Furthermore, this comparison helps understand the tradeoffs between different metrics and assists in selecting the most appropriate evaluation measures for specific detection tasks. Specifically, YOLOv5 and our proposed method were implemented utilizing ultralytics.[1] The

[1]https://github.com/ultralytics/yolov5

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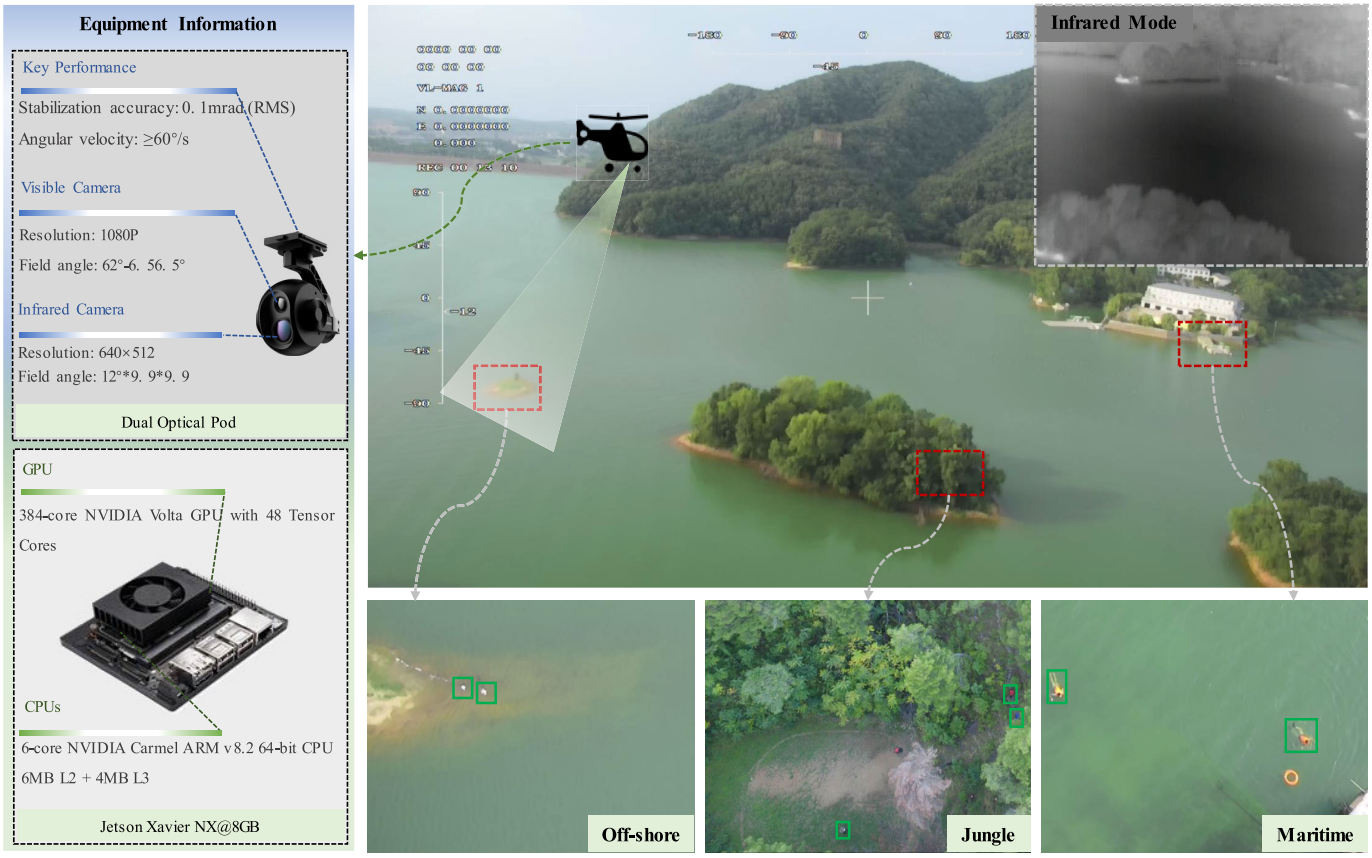



Fig. 7. Illustration of the collection and processing procedures employed by the SaR system. The scenario discussed originates from a scenic area in Xinyang, China. In this particular case, samples were collected, encompassing both RGB and infrared bands, ultimately creating the VTSaR dataset.

remaining methods (see Tables I and II) were implemented with mmdetection.[2] It is important to note that these methods were trained using the same initialization parameters.

### A. Implementation Details

*1) Parameters Setting:* Experiments are conducted on a Ubuntu 22.04.1 LTS (64-bit) operating system, running on an Intel Core i9-13900KF processor with 6 cores and 32 GB RAM, clocked at 5.80 GHz. GPU server is also aided by an NVIDIA GeForce RTX 4090 GPU with 24 GB memory. Models are trained for 100 epochs using the stochastic gradient descent (SGD) optimizer, set with a momentum of 0.9 and weight decay of 0.0001. The initial learning rate is 0.01 and decayed at epoch 10. Our proposed method undergoes testing and validation on three datasets: Heridal [35], MUPD dataset [36], and VTSaR. These datasets were chosen due to the presence of small objects, defined as objects smaller than 32 pixels according to the COCO definition. The descriptions of the datasets are provided below.

1) Heridal dataset consists of 1650 high-resolution images (4000 × 3000) captured by uncrewed helicopters, portraying individuals in nonurban settings such as mountains, forests, oceans, and deserts.
2) MUPD dataset comprises 33 videos captured by two UAVs flying at 10–50 m, containing 13 462 sampled images with bounding box annotations for 153 112 person objects.
3) VTSaR dataset offers diverse scenes depicting urban areas, seas, shores, and jungles. Samples were collected from various locations using our custom-built unmanned helicopter equipped with a dual-camera gimbal system (see Fig. 7). These samples comprise aerial scenarios diverse in geography, time, and environmental conditions. To ensure variety and representativeness, we chose images depicting persons engaging in various activities, like standing, walking, and working. In the dataset annotation stage, preliminary screening processes were used to discard images of blurriness or obstructions. Thereafter, persons in the aerial images were identified, and their positions were annotated using bounding boxes. VTSaR comprises 17 video clips, 41 269 frames, and 4099 still images, all annotated with single labels for aligned dual-modality images.
4) VisDrone dataset stands as a vast repository of UAVs video data, integral to computer vision and object detection research. This dataset comprises 288 video clips totaling approximately 10 h of footage. It encompasses 261 908 videos and 10 209 frames of static imagery, annotated for 2 806 154 objects and 7 663 482 object instances. The dataset offers an extensive geographical display, spanning diverse locations across 14 disparate cities in China, covering thousands of kilometers.

[2]https://github.com/open-mmlab/mmdetection

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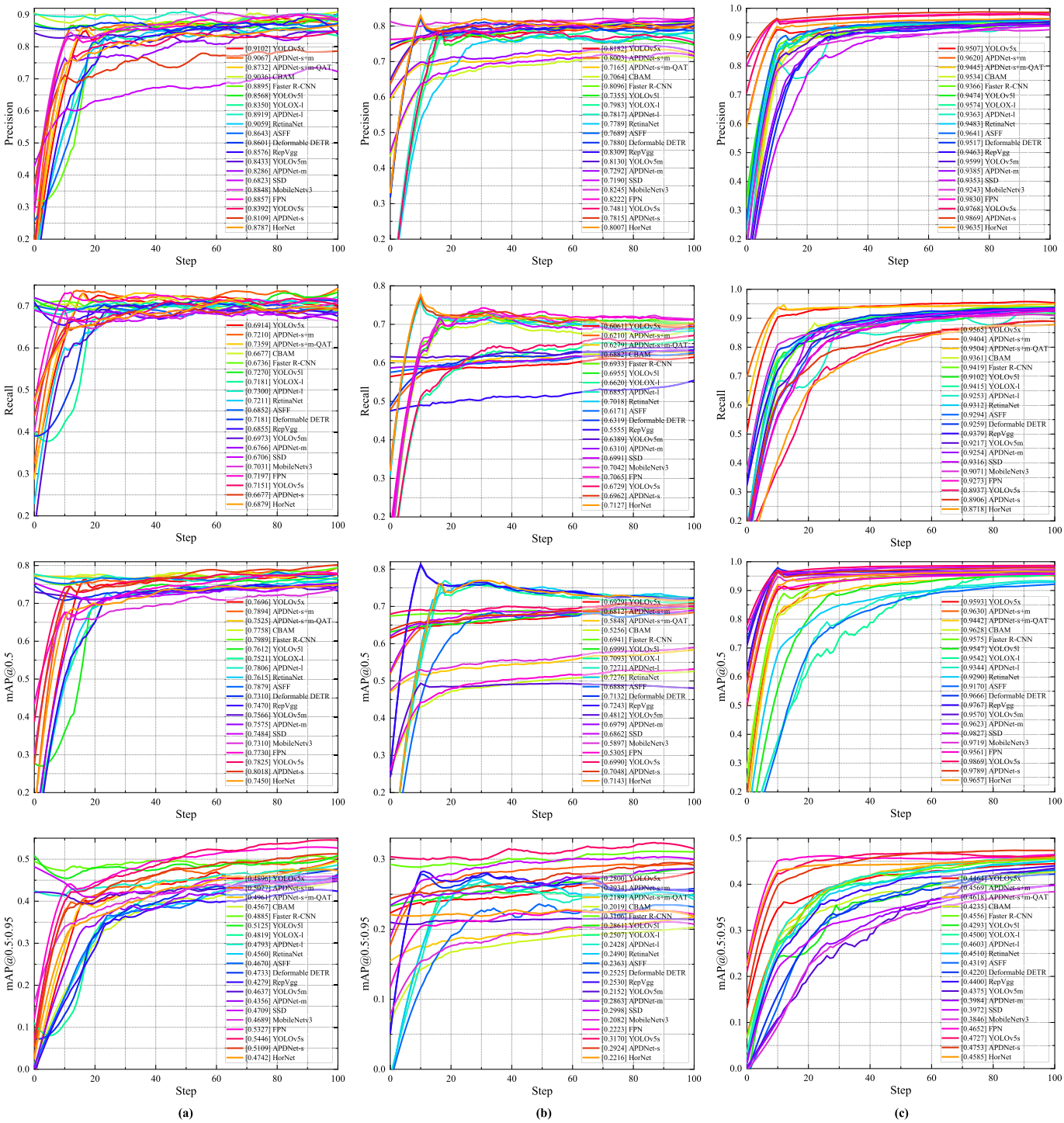



Fig. 8. Comparative analysis of detector training curves on the three datasets: (a) Heridal, (b) MUPD, and (c) VTSaR. Distinct methods or models are depicted with varying colors, and it is crucial to observe that the scaling on the vertical axis is not entirely consistent.

The SaR system utilized in this study consists of a dual-source acquisition gimbal, an embedded development board, communication devices, and additional components, as illustrated in Fig. 7. The scenario is derived from the VTSaR dataset and was collected in a scenic area located in Xinyang, China. Using an uncrewed helicopter facilitates access to search areas that are typically inaccessible to personnel. This enables the execution of detection and monitoring tasks effectively.

*2) Evaluation Criteria:* To assess the detection performance of our proposed method, we select some training accuracy metrics, precision, recall, mAP@0.5, and mAP@0.95, as well as some inference speed metrics, parameters, GFLOPs, Letency, and frames per second (FPS) as the evaluation criteria. The descriptions of the metrics are provided below.

Precision captures the ratio of correctly detected objects to the total number of detected objects. Recall measures the ratio between the number of correctly detected objects

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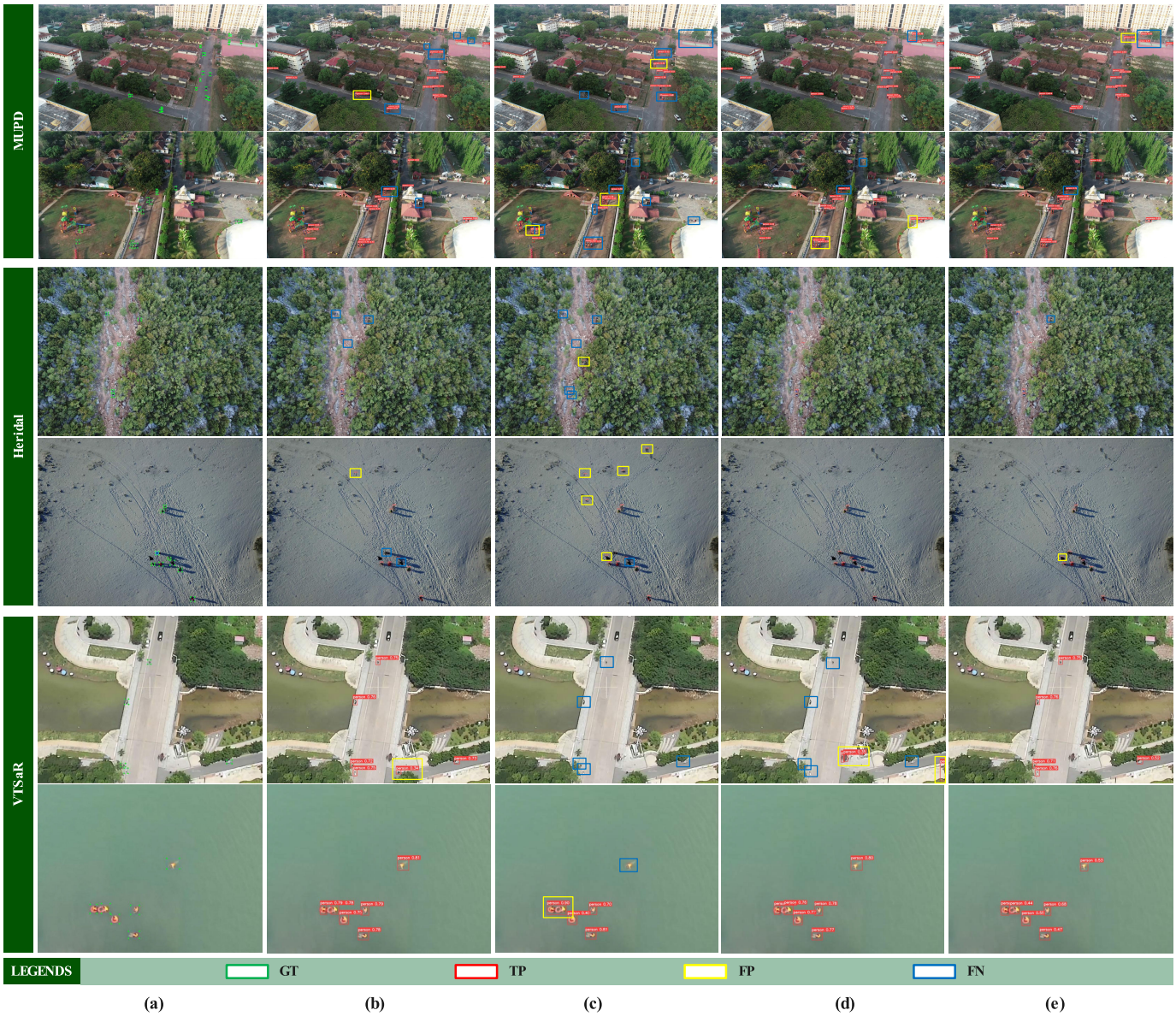



Fig. 9. Statistical analysis of lightweight evaluation metrics for detection models. GT, TP, FP, and FN are represented by color blocks: green, red, yellow, and blue, respectively. (a) GT. (b) Faster R-CNN. (c) Deformable DETR. (d) RetinaNet. (e) APDNet (Ours).

and the total number of actual objects. mAP@0.5 represents the average precision with an IoU threshold above 0.5. mAP@0.5:0.95 denotes average precision within an IoU threshold ranging from 0.5 to 0.95, with a step size of 0.05. Parameters correspond to the count of trainable parameters within a network, typically including weights (W) and biases (b). GFLOPs denote the measurement of floating-point operations per second, where addition, subtraction, multiplication, or division operations each count as a single FLOP. Latency refers to the time required for model inference and postprocessing, with lower values indicating faster execution speed. FPS represents the number of images a CNN can process in one second.

### *B. Comparison of Detection Accuracy Metrics*

*1) Heridal Dataset:* Considering the high resolution of samples in the Heridal dataset (4000 × 3000), directly using the default resolution in the detector may lead to memory overflow. Therefore, the original images were resized using the letterbox strategy to 1536 × 1536. Fig. 8(a) provides an overview of the training metrics for different algorithms on the Heridal dataset. Some models underwent fine-tuning instead of training from scratch due to convergence issues or suboptimal hyperparameter settings. Initially, the YOLOv5s model was trained for 100 epochs to obtain a pretrained model, which was then further trained on different models to acquire the metrics. Table I presents the quantitative analysis of training metrics for various models. Results show that CBAM achieves the highest precision metric at 93.67%, FPN achieves the highest recall metric at 76.19%, Faster R-CNN achieves the highest mAP@0.5 metric at 80.43%, and YOLOv5s achieves the highest mAP@0.5:0.95 metric at 54.94%. Furthermore, our proposed method consistently exhibits a minor decrease in training accuracy metrics, with the largest drop observed

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TABLE I

COMPARATIVE ANALYSIS OF TRAINING ACCURACY METRICS ACROSS THREE DATASETS WITH DIFFERENT ALGORITHMS. THE OPTIMAL VALUES ARE DENOTED IN BOLD

| *Method @ RTX 4090x12GB* | | **Heridal @ 1536x1536** | | | | **MUPD @ 1280x1280** | | | | **VTSaR @ 640x640** | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | *Precision* ($10^{-2}$) | *Recall* ($10^{-2}$) | *mAP@0.5* ($10^{-2}$) | *mAP@0.5:0.95* ($10^{-2}$) | *Precision* ($10^{-2}$) | *Recall* ($10^{-2}$) | *mAP@0.5* ($10^{-2}$) | *mAP@0.5:0.95* ($10^{-2}$) | *Precision* ($10^{-2}$) | *Recall* ($10^{-2}$) | *mAP@0.5* ($10^{-2}$) | *mAP@0.5:0.95* ($10^{-2}$) |
| CBAM [49] | | **93.67** | 74.78 | 79.42 | 45.9 | 72.57 | 73.79 | 52.87 | 20.31 | 95.6 | 94.69 | 96.49 | 42.98 |
| Faster R-CNN [50] | | 92.88 | 73.89 | **80.43** | 51.9 | 82.23 | 73.92 | 70.27 | 31.33 | 95.49 | 94.2 | 95.84 | 45.57 |
| YOLOX-l [51] | | 88.22 | 73.89 | 77.34 | 48.39 | 80.99 | 68.47 | 77.15 | 27.45 | 96.14 | 94.2 | 95.51 | 45.02 |
| RetinaNet [52] | | 92.16 | 74.19 | 78.13 | 47.14 | 81.71 | 74.08 | 78.15 | 28.78 | 96.33 | 94.06 | 93.05 | 45.11 |
| ASFF [53] | | 92.19 | 72.71 | 79.72 | 46.71 | 84.19 | 63.89 | 71.5 | 26.49 | 96.56 | 93.17 | 92.47 | 43.22 |
| Deformable DETR [54] | | 92.52 | 74.19 | 77.62 | 48.75 | 83.08 | 64.84 | 77.97 | 28.95 | 95.42 | 94.19 | 97.14 | 42.21 |
| RepVgg [55] | | 93.66 | 72.69 | 76.19 | 44.3 | 83.56 | 56.11 | 78.85 | 29.43 | 94.79 | 94.05 | 97.68 | 44.04 |
| SSD [56] | | 75.22 | 71.8 | 75.78 | 48.02 | 74.05 | 74.7 | 70.72 | 31.14 | 94.38 | 93.48 | 98.31 | 39.73 |
| MobileNetv3 [57] | | 92.87 | 72.11 | 74.92 | 47.33 | 83.74 | 75.88 | 59.85 | 21.83 | 93.37 | 91.92 | 97.34 | 38.47 |
| FPN [58] | | 92.17 | **76.19** | 78.96 | 53.4 | 83.42 | **76.73** | 53.86 | 23.68 | 98.48 | 92.74 | 95.8 | 46.52 |
| HorNet [59] | | 87.94 | 74.19 | 75.4 | 48.4 | **84.85** | 75.9 | **79.13** | 23.57 | 96.53 | 89.1 | 96.66 | 45.88 |
| YOLOv5 [9] | YOLOv5s | 86.12 | 73.5 | 78.63 | **54.94** | 81.85 | 69.36 | 71.65 | **32.97** | 98.1 | 89.43 | **98.71** | 47.27 |
| | YOLOv5m | 88.75 | 72.71 | 76.53 | 46.58 | 83.78 | 65.3 | 50.03 | 21.74 | 96.08 | 93.17 | 96.16 | 43.75 |
| | YOLOv5l | 90.9 | 75.97 | 77.67 | 51.26 | 80.51 | 74.76 | 70.27 | 28.8 | 95.92 | 92.63 | 95.49 | 43.33 |
| | YOLOv5x | 91.74 | 74.78 | 79.28 | 50.64 | 82.06 | 62.82 | 69.77 | 28.24 | 96.74 | **95.98** | 96.51 | 44.64 |
| Ours | APDNet-s | 81.1 | 71.82 | 80.23 | 51.55 | 84.61 | 75.32 | 71.67 | 30.15 | **98.86** | 89.58 | 97.93 | **47.58** |
| | APDNet-m | 88.65 | 72.41 | 76.24 | 46.46 | 76.76 | 64.78 | 70.76 | 29.71 | 95.01 | 92.88 | 96.32 | 39.85 |
| | APDNet-l | 93.4 | 73.6 | 80.15 | 49.33 | 81.35 | 74.14 | 77.36 | 28.63 | 94.7 | 92.68 | 93.5 | 46.04 |
| | APDNet-s+m | 91.27 | 76.56 | 80.38 | 51.31 | 82.3 | 63.07 | 68.8 | 29.51 | 96.51 | 94.86 | 96.57 | 45.72 |
| | APDNet-s+m-QAT | 92.49 | 75.08 | 80.26 | 50.35 | 75.43 | 63.64 | 59 | 22.16 | 94.97 | 95.04 | 95.57 | 46.19 |

in mAP@0.5:0.95 at 4.59%. This can be attributed to our lightweight architecture and weight quantization, resulting in a slight performance reduction within an acceptable range of 0%–5%. It is worth noting that all models have relatively low recall metrics, which may be attributed to the majority of samples in the Heridal dataset being collected from mountainous and jungle regions, where instances of persons are relatively scarce. This is a matter that warrants further ongoing research.

*2) MUPD Dataset:* The MUPD dataset includes GT annotations in YOLO and JSON formats, with data samples partitioned into train, validation, and test sets. For consistency, the dataset configuration was maintained as per the original authors' specifications during algorithm testing. Fig. 8(b) shows the training metrics of different algorithms on the MUPD dataset. Similar to the training strategy employed for the Heridal dataset, some models underwent fine-tuning rather than starting training from scratch. Specifically, the YOLOv5s model underwent 100 epochs of training to obtain a pretrained model, which was then utilized for subsequent training on various models to obtain the metric results. Table I presents a quantitative analysis of the training metrics for different models. Performance analysis reveals that HorNet achieves the highest precision metric at 84.85%, while FPN achieves the highest recall metric at 76.73%. Moreover, HorNet achieves the highest mAP@0.5 metric at 79.31%, and YOLOv5s achieves the highest mAP@0.5:0.95 metric at 32.97%. Compared to the methods proposed in MUPD, our proposed method APDNet-s showcases a notable performance improvement of 10.17%. It sets a benchmark for precision, recall, and mAP@0.5:0.95. However, subsequent testing to evaluate the effectiveness of knowledge distillation and quantization on this dataset revealed two key findings. First, all training metrics experienced a significant decline, with a maximum drop of mAP@0.5 = 20.13% compared to the best training results. Second, conducting distillation alone resulted in a minimal drop of only mAP@0.5 = 10.33%. These results demonstrate that the impact of model quantization can differ based on the dataset, thereby emphasizing the need to assess the robustness of our methodology across various datasets, given that its efficacy may vary.

*3) VTSaR Dataset:* The VTSaR dataset, as depicted in Fig. 6, comprises dual-source data: visible light and infrared, which are stored in a picture-in-picture format with dimensions of 1920 × 1080, facilitating image registration. However, when training with single-source data, infrared images tend to interfere with the model training. Hence, we adopted an image cropping approach to separately extract the visible light and infrared regions as images with dimensions of 640 × 480. The training metrics of different algorithms on the VTSaR dataset are presented in Fig. 8(c). In contrast to the Heridal and MUPD datasets, all models were trained from scratch on the VTSaR dataset. This decision was made due to the relatively small size of the image data and the capability of GPU performance to handle input sizes for all models in a single batch. Table I provides a quantitative analysis of the training metrics for various models. It can be observed that APDNet-s achieves the highest precision metric at 98.86%, YOLOv5x attains the highest recall metric at 95.98%, YOLOv5s achieves the highest mAP@0.5 metric at 98.71%, and APDNet-s achieves the highest mAP@0.5:0.95 metric at 47.58%. However, without exception, our proposed method exhibits a decrease in training accuracy metrics compared to the optimal results, with the maximum drop reaching 3.89% in precision. This decrease can be attributed to the lightweight architecture and quantization undergone by our model, resulting in some information loss. Nevertheless, all the drops remain within an acceptable range

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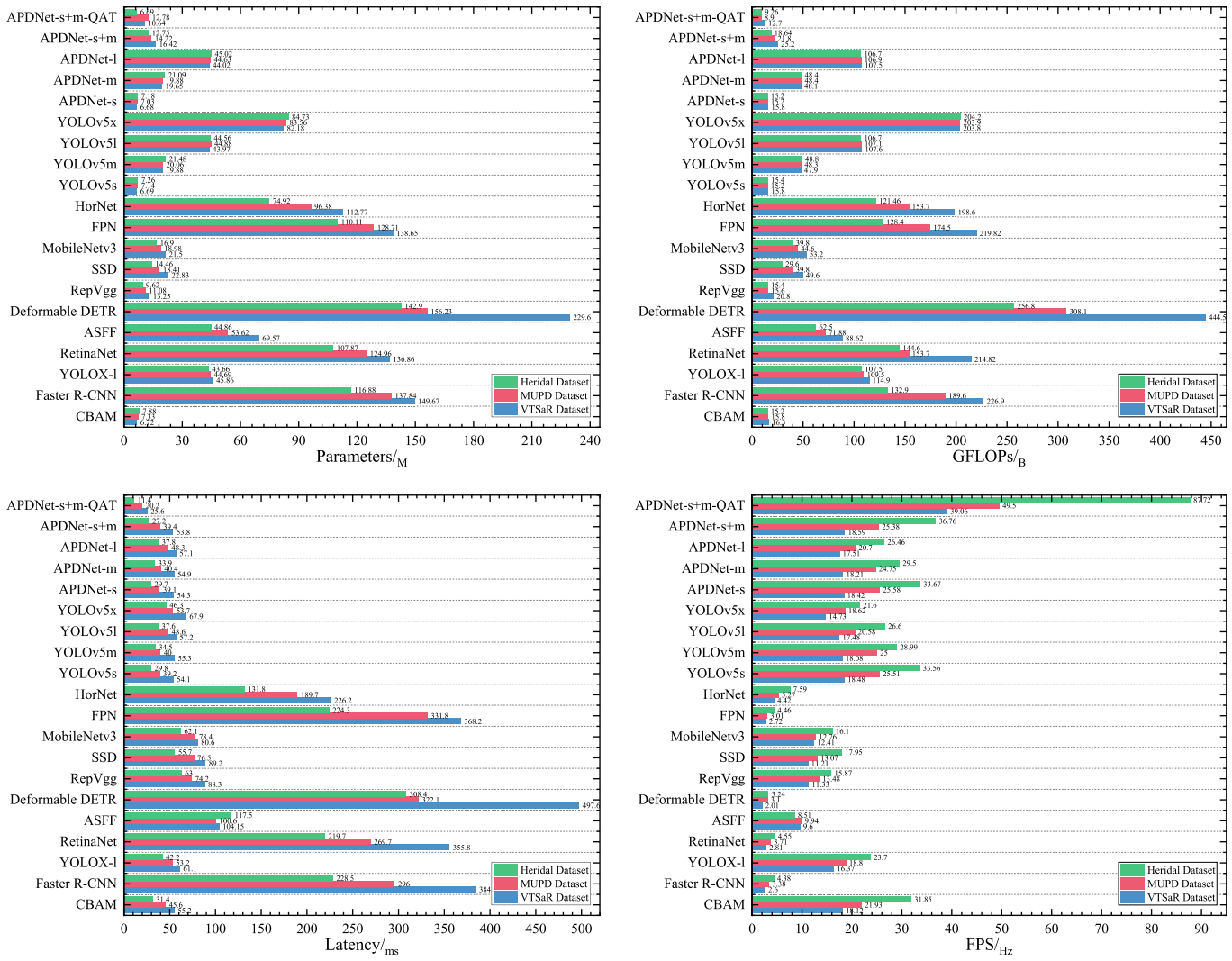



Fig. 10. Visualization of APD: a comparison between SOTA methods and APDNet (Ours). Various detection errors have been meticulously annotated using rectangular bounding boxes in distinct colors.

(typically 0%–5% for lightweight models). Furthermore, the superior performance of the metrics on the VTSaR dataset can be attributed to the cropping strategy that we applied during both training and validation.

In conclusion, after conducting multiple tests and comparative analyses on three distinct datasets, the results have validated our proposed method's inherent robustness and the permissible extent of accuracy metric decline.

### C. Comparison of Detection Inference Speed

Achieving a balance between accuracy and speed in the detection model is paramount. Table II presents the lightweight evaluation metrics for various algorithms, as discussed in Section IV-B, across three datasets. All model tests were conducted using the trained models detailed in Section IV-B and are classified into our method (five types), YOLOv5 (four types), and other methods (11 types). The insights garnered from Table II are as follows: 1) the mAP@0.5 metric experiences a maximum decline of 1.77% when compared to the highest accuracy metric. 2) Our approach demonstrates significant improvement across all lightweight evaluation metrics, thereby meeting the real-time requirements of the RTX4090. 3) The resolution of input images substantially impacts lightweight evaluation metrics, sometimes resulting in exponential decreases in FPS. 4) Compared to the baseline YOLOv5, our method shows minor fluctuations in accuracy metrics but significantly surpasses YOLOv5 in inference speed. Fig. 9 provides a qualitative analysis of the lightweight evaluation metrics, illustrating the model's parameter count, GFLOPs, latency, and FPS across the three datasets. This visualization offers an intuitive depiction of the metric distribution across the datasets and guides future research on lightweight methods.

Consequently, Fig. 10 depicts the visualization of detection results across the three datasets. Each pair of rows corresponds to a dataset's test samples, while each column showcases the GT and the testing outcomes of various algorithms. Due to the objects' diminutive nature, erroneous results were manually labeled. The visualized results demonstrate that while APDNet does not attain the finest detection performance among the three datasets, our foremost objective centers on striking a harmonious equilibrium between accuracy and speed (which

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TABLE II

COMPARATIVE ANALYSIS OF LIGHTWEIGHT EVALUATION METRICS ACROSS THREE DATASETS FOR DIFFERENT ALGORITHMS. THE OPTIMAL VALUES ARE HIGHLIGHTED IN BOLD

| *Method @RTX 4090x12GB* | | **Heridal @1536x1536** | | | | | **MUPD @1280x1280** | | | | | **VTSaR @640x640** | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | *mAP@0.5 ($10^{-2}$)* | *Parameters (M)* | *GFLOPs (B)* | *Latency (ms)* | *FPS (Hz)* | *mAP@0.5 ($10^{-2}$)* | *Parameters (M)* | *GFLOPs (B)* | *Latency (ms)* | *FPS (Hz)* | *mAP@0.5 ($10^{-2}$)* | *Parameters (M)* | *GFLOPs (B)* | *Latency (ms)* | *FPS (Hz)* |
| CBAM [49] | | 79.42 | 6.72 | 16.3 | 55.2 | 18.12 | 52.87 | 7.33 | 15.8 | 45.6 | 21.93 | 96.49 | 7.88 | 15.2 | 31.4 | 31.85 |
| Faster R-CNN [50] | | **80.43** | **149.67** | 226.9 | 384.5 | 2.6 | 70.27 | 137.84 | 189.6 | 296 | 3.38 | 95.84 | **116.88** | 132.9 | 228.5 | 4.38 |
| YOLOX-l [51] | | 77.34 | 45.86 | 114.9 | 61.1 | 16.37 | 77.15 | 44.69 | 109.5 | 53.2 | 18.8 | 95.51 | 43.66 | 107.5 | 42.2 | 23.7 |
| RetinaNet [52] | | 78.13 | 136.86 | 214.82 | 355.8 | 2.81 | 78.15 | 124.96 | 153.7 | 269.7 | 3.71 | 93.05 | 107.87 | 144.6 | 219.7 | 4.55 |
| ASFF [53] | | 79.72 | 69.57 | 88.62 | 104.15 | 9.6 | 71.5 | 53.62 | 71.88 | 100.6 | 9.94 | 92.47 | 44.86 | 62.5 | 117.5 | 8.51 |
| Deformable DETR [54] | | 77.62 | 229.6 | **444.5** | **497.6** | **2.01** | 77.97 | **156.23** | **308.1** | 322.1 | 3.1 | 97.14 | 142.9 | **256.8** | **308.4** | **3.24** |
| RepVgg [55] | | 76.19 | 13.25 | 20.8 | 88.3 | 11.33 | 78.85 | 11.08 | 15.6 | 74.2 | 13.48 | 97.68 | 9.62 | 15.4 | 63 | 15.87 |
| SSD [56] | | 75.78 | 22.83 | 49.6 | 89.2 | 11.21 | 70.72 | 18.41 | 39.8 | 76.5 | 13.07 | 98.31 | 14.46 | 29.6 | 55.7 | 17.95 |
| MobileNetv3 [57] | | 74.92 | 21.5 | 53.2 | 80.6 | 12.41 | 59.85 | 18.98 | 44.6 | 78.4 | 12.76 | 97.34 | 16.9 | 39.8 | 62.1 | 16.1 |
| FPN [58] | | 78.96 | 138.65 | 219.82 | 368.2 | 2.72 | 53.86 | 128.71 | 174.5 | **331.8** | **3.01** | 95.8 | 110.11 | 128.4 | 224.3 | 4.46 |
| HorNet [59] | | 75.4 | 112.77 | 198.6 | 226.2 | 4.42 | **79.13** | 96.38 | 153.7 | 189.7 | 5.27 | 96.66 | 74.92 | 121.46 | 131.8 | 7.59 |
| YOLOv5 [9] | YOLOv5s | 78.63 | 6.69 | 15.8 | 54.1 | 18.48 | 71.65 | 7.14 | 15.2 | 39.2 | 25.51 | **98.71** | 7.26 | 15.4 | 29.8 | 33.56 |
| | YOLOv5m | 76.53 | 19.88 | 47.9 | 55.3 | 18.08 | 50.03 | 20.06 | 48.3 | 40 | 25 | 96.16 | 21.48 | 48.8 | 34.5 | 28.99 |
| | YOLOv5l | 77.67 | 43.97 | 107.6 | 57.2 | 17.48 | 70.27 | 44.88 | 107.1 | 48.6 | 20.58 | 95.49 | 44.56 | 106.7 | 37.6 | 26.6 |
| | YOLOv5x | 79.28 | 82.18 | 203.8 | 67.9 | 14.73 | 69.77 | 83.56 | 203.9 | 53.7 | 18.62 | 96.51 | 84.73 | 204.2 | 46.3 | 21.6 |
| Ours | APDNet-s | 80.23 | **6.68** | 15.8 | 54.3 | 18.42 | 71.67 | **7.03** | 15.2 | 39.1 | 25.58 | 97.93 | 7.18 | 15.2 | 29.7 | 33.67 |
| | APDNet-m | 76.24 | 19.65 | 48.1 | 54.9 | 18.21 | 70.76 | 19.88 | 48.4 | 40.4 | 24.75 | 96.32 | 21.09 | 48.4 | 33.9 | 29.5 |
| | APDNet-l | 80.15 | 44.02 | 107.5 | 57.1 | 17.51 | **77.36** | 44.63 | 106.9 | 48.3 | 20.7 | 93.5 | 45.02 | 106.7 | 37.8 | 26.46 |
| | APDNet-s+m | **80.38** | 16.42 | 25.2 | 53.8 | 18.59 | 68.8 | 14.22 | 21.8 | 39.4 | 25.38 | 96.57 | 12.75 | 18.64 | 27.2 | 36.76 |
| | APDNet-s+m-QAT | 80.26 | 10.64 | **12.7** | **25.6** | **39.06** | 59 | 12.78 | **8.9** | **20.2** | **49.5** | 95.57 | **6.69** | **9.26** | **11.4** | **87.72** |

TABLE III

COMPARATIVE ANALYSIS OF DIFFERENT MODULES WITH APDNET. KEY INDICATORS ARE VISUALLY REPRESENTED USING COLOR CODING, WITH OPTIMAL mAP@0.5 VALUES MARKED IN BLUE AND OPTIMAL FPS VALUES IN RED

| **Method** | **Module** | | **Heridal @1536x1536** | | | | | **MUPD @1280x1280** | | | | | **VTSaR @640x640** | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | *TAD* | *LPQ* | *mAP@0.5 ($10^{-2}$)* | *Parameters (M)* | *GFLOPs (B)* | *Latency (ms)* | *FPS (Hz)* | *mAP@0.5 ($10^{-2}$)* | *Parameters (M)* | *GFLOPs (B)* | *Latency (ms)* | *FPS (Hz)* | *mAP@0.5 ($10^{-2}$)* | *Parameters (M)* | *GFLOPs (B)* | *Latency (ms)* | *FPS (Hz)* |
| Baseline | ✗ | ✗ | 78.63 | 6.69 | 15.8 | 54.1 | 18.48 | 71.65 | 7.14 | 15.2 | 39.2 | 25.51 | 98.71 | 7.26 | 15.4 | 29.8 | 33.56 |
| Ours | ✓/s+m | ✗ | 80.38 | 16.42 | 25.2 | 53.8 | 18.59 | 68.8 | 14.22 | 21.8 | 39.4 | 25.38 | 96.57 | 12.75 | 18.64 | 27.2 | 36.76 |
| | ✗ | ✓ | 76.44 | 4.28 | 6.4 | 38.7 | 25.84 | 68.19 | 6.55 | 6.3 | 18.4 | 54.35 | 96.13 | 5.07 | 6.6 | 17.1 | 58.48 |
| | ✓/s+m | ✓ | 80.26 | 10.64 | 12.7 | 25.6 | 39.06 | 59 | 12.78 | 8.9 | 20.2 | 49.5 | 95.57 | 6.69 | 9.26 | 11.4 | 87.72 |




Fig. 11. Visualization of tracking-assisted detection results at two sites under Jetson Xavier NX platform.

can be analyzed in conjunction with Fig. 9) rather than exclusively optimizing the accuracy metric.

## D. Ablation Study

In this section, we performed a series of ablation experiments to assess the impact and significance of each component in our proposed model. Specifically, we systematically removed or modified distinct parts of the model, enabling us to analyze the resulting changes in its performance.

*1) Effect of TAD:* A model's performance can be enhanced by transferring response information from a teacher to a student model using response-based knowledge distillation. To evaluate the impact of knowledge distillation on detector performance, we investigated the activation or freezing of the TAD module. The baseline model employed YOLOv5s, while the APDNet utilized a pretrained model adapted from YOLOv5m to YOLOv5s. The impact of knowledge distillation on the APDNet detector's training metrics, evaluated using mAP@0.5, relative to the baseline, is presented in Table III.

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TABLE IV
DETECTION PERFORMANCE METRICS OF APDNET ON DIFFERENT EDGE DEVICES. THE VALUES IN EACH CELL REPRESENT THE MEANING AS FOLLOWS: MAP@0.5/FPS. THE OPTIMAL VALUES FOR EACH ARE MARKED, RESPECTIVELY, IN BLUE AND RED

| *Device Info* | *Tracker & Detector* | *YOLOv5-s* | *YOLOv5-m* | *APDNet-s+m* | *APDNet-s+m & QAT* |
|---|---|---|---|---|---|
| *RTX 4090 @24GB* | KCF [61] | 95.48 / 2.26 | 91.67 / 1.24 | 90.85 / 0.69 | 92.31 / 2.77 |
| | Sort [62] | 98.27 / 9.58 | 97.19 / 8.25 | 97.57 / 7.46 | 95.06 / 14.32 |
| | DeepSort [63] | 98.02 / 6.31 | 97.56 / 5.17 | 97.59 / 3.75 | 93.71 / 8.43 |
| | StrongSort [64] | 98.89 / 7.83 | 97.16 / 5.84 | 98 / 6.52 | 95.44 / 11.15 |
| | ByteTrack [65] | 98.84 / 21.42 | 97.25 / 15.6 | 97.18 / 17.44 | 95.72 / 35.26 |
| *Jetson Xavier NX@8GB* | KCF [61] | 92.5 / 1.02 | 88.63 / 0.75 | 89.14 / 0.08 | 91.58 / 1.37 |
| | Sort [62] | 96.72 / 4.76 | 96.88 / 3.96 | 97.01 / 3.59 | 94.28 / 7.85 |
| | DeepSort [63] | 95.45 / 1.7 | 95.82 / 1.23 | 96.23 / 1.14 | 92.15 / 2.81 |
| | StrongSort [64] | 96.58 / 3.62 | 95.49 / 3.77 | 97.43 / 2 | 94.56 / 4.81 |
| | ByteTrack [65] | 96.37 / 15.36 | 96.49 / 7.62 | 96.08 / 9.49 | 94.86 / 22.84 |

The results indicate achievement of 80.38% (1.75↑), 68.8% (2.85↓), and 96.57% (2.14↓) on the three benchmark datasets, respectively. Notably, the inference speed remained comparable to that of the baseline model.

*2) Effect of LPQ:* Quantization is a commonly used method for optimizing models by substantially reducing their storage requirements and computational complexity while maintaining a satisfactory level of performance. To evaluate the impact of quantization on detector performance, we investigated the activation or freezing of the LPQ module. The experimental results presented in Table III demonstrate the changes in inference FPS for the APDNet detector across three benchmark datasets relative to the baseline: 25.84 (7.63↑), 54.35 (28.84↑), and 58.48 (24.92↑). Notably, the mAP@0.5 remained comparable to that of the baseline model.

### E. Applications

This section aims to facilitate the robust deployment of APD applications by combining the lightweight detection model with a plug-and-play tracker. We conducted a comparative analysis of different tracking-assisted detection methods on RTX 4090 and Jetson Xavier NX platforms to achieve this goal. Table IV presents the evaluation metrics using mAP@0.5/FPS as the representative measurement. Our method achieved the optimal balance on the VTSaR dataset for both the GPU server 4090 and the embedded device Jetson Xavier NX, with mAP@0.5/FPS values of 95.72%/35.26 and 94.86%/22.84, respectively. These results meet real-time requirements while effectively handling the motion and variations of the targets. Fig. 11 provides a qualitative analysis of APDNet-s+m-QAT using the ByteTrack method in two scenarios (tested on Jetson Xavier NX), showcasing scene examples extracted from the VisDrone [60] and Heridal datasets. The image resolutions for these scenarios are $1920 \times 1080$ and $4000 \times 3000$, respectively.

## V. CONCLUSION

In this study, we investigated the effectiveness of knowledge distillation and quantization techniques in APD. Through extensive experiments and analysis, we have made several noteworthy discoveries. First, we demonstrated that the TAD module can efficiently transfer expertise from a teacher to a student model, enhancing APD performance. By distilling knowledge from a larger and more complex model into a smaller and lighter model, we struck a balance between detection accuracy and computational efficiency. Second, we explored the advantages of the LPQ module in reducing computational complexity and model size. On the three datasets, Heridal, MUPD, and VTSaR, APDNet achieved optimal mAP@0.5 scores of 80.38%, 77.36%, and 97.93%, respectively. Despite only minor performance losses of 0.05%, 1.77%, and 0.78% compared to the most advanced algorithms, the speed metric FPS increased significantly to 39.06, 49.5, and 87.72, satisfying real-time requirements. In conclusion, utilizing knowledge distillation and quantization techniques improved detection performance and computational efficiency in APD. The implications of this research are significant for the practical deployment of aerial surveillance systems. As avenues for future research, we recommend exploring the application of these techniques on more challenging datasets, investigating different approaches for image generation [66], and fusion detection in edge devices, and analyzing the trade-offs between model compression and detection accuracy.

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**Xiangqing Zhang** (Graduate Student Member, IEEE) received the M.S. degree in information and communication engineering from Chang'an University, Xi'an, China, in 2018. He is currently pursuing the Ph.D. degree with the School of Electronics and Information, Northwestern Polytechnical University, Xi'an.

Meanwhile, he is a Lecturer with the College of Mathematics and Computer Science, Yan'an University, Yan'an, China. His research interests include deep learning, data mining, and object detection of remote sensing images.



**Yan Feng** (Member, IEEE) received the B.S., M.S., and Ph.D. degrees from the School of Electronics and Information Engineering, Northwestern Polytechnical University, Xi'an, China, in 1984, 1989, and 2007, respectively.

In 2008, she was a Visiting Scholar with the Department of Electrical Engineering, The University of Adelaide, Adelaide, SA, Australia. She is currently a Professor with the School of Electronics and Information Engineering, Northwestern Polytechnical University. Her research interests include image processing, hyperspectral image classification, compressing and recognition, object detection and tracking, noise reduction, compressed sensing theory, and application.



**Shun Zhang** (Member, IEEE) received the B.S. and Ph.D. degrees in electronic engineering from Xi'an Jiaotong University, Xi'an, China, in 2009 and 2016, respectively.

He is currently an Associate Professor with the School of Electronics and Information, Northwestern Polytechnical University, Xi'an. His research interests include machine learning, computer vision, and human–computer interaction, with a focus on object detection, visual tracking, person reidentification, image classification, feature extraction, and sparse representation.



**Nan Wang** (Graduate Student Member, IEEE) received the M.S. degree from the College of Computer Science and Technology, Xidian University, Xi'an, China, in 2017. He is currently pursuing the Ph.D. degree in information and communication engineering with Northwestern Polytechnical University, Xi'an.

His research interests include deep learning, data mining, super-resolution, and object detection of remote sensing images.



**Guohua Lu** received the B.S. and M.S. degrees in biomedical engineering and the Ph.D. degree in biophysics from Fourth Military Medical University, Xi'an, China, in 1999, 2002, and 2005, respectively.

He was a Visiting Scholar with the Department of Physiology, University of Oxford, Oxford, U.K., from September 2007 to May 2009. He is currently a Professor with the Department of Military Biomedical Engineering, Fourth Military Medical University. His research interests include remote sensing image processing and applications, intelligent signal and information acquisition, and object detection and tracking.



**Shaohui Mei** (Senior Member, IEEE) received the B.S. degree in electronics and information engineering and the Ph.D. degree in signal and information processing from Northwestern Polytechnical University, Xi'an, China, in 2005 and 2011, respectively.

He was a Visiting Student at The University of Sydney, Camperdown, NSW, Australia, from October 2007 to October 2008. He is currently a Professor with the School of Electronics and Information, Northwestern Polytechnical University. His research interests include hyperspectral remote sensing image processing and applications, intelligent signal and information acquisition and processing, video processing, and pattern recognition.

Dr. Mei was a recipient of the Excellent Doctoral Dissertation Award of Shaanxi Province in 2014 and the Best Paper Award of the IEEE International Symposium on Intelligent Signal Processing and Communication Systems (ISPACS) in 2017. He was awarded the Best Reviewer of the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING in 2022 and IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING in 2019. He is serving as a Topical Associate Editor for IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, an Associate Editor for IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING, and a reviewer for more than 20 international famous academic journals.

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